

Daily Algebra Drill #6

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Comparing coefficients · Substitution $t = \sqrt{x}$ for surd equations · Schur's inequality · Integer-valued polynomials.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

No expansion. Recognise the pattern, write the answer.

- Factorise: $x^3 + 5x^2 + 6x$.
- Given $x + \frac{1}{x} = \sqrt{5}$, find $x^2 + \frac{1}{x^2}$.
- Without expanding, state the coefficient of xy^2 in $(x + y)^3$.
- Evaluate: $\frac{1}{\sqrt{2} - 1} - \frac{1}{\sqrt{2} + 1}$.
- The polynomial $f(x) = x^3 + ax + b$ has $x = 2$ as a root and $f'(2) = 0$... but no calculus today! Instead: if $x = 2$ is a *repeated* root, find a and b .
- Simplify: $\left(\frac{2x}{y}\right)^3 \div \frac{4x^2}{y^2}$.
- Given $\alpha\beta = 6$ and $\alpha + \beta = 5$, find $\alpha^2\beta + \alpha\beta^2$.
- Factorise over $\mathbb{Z}$: $4a^2 - 4ab + b^2$.
- Show quickly that $2^{10} = 1024$, hence simplify $2^{10} - 1$ as a product of three integer factors greater than 1.
- Given $f(x) = x^2 - 3x + 2$, find $f(f(0))$.

Section B Manipulation Drills

(≤50 s each)

Fractions, structure, identities. Minimum working.

- Find constants A, B, C such that $x^2 + 3x + 5 \equiv A(x + 1)^2 + B(x + 1) + C$.
- The polynomial $p(x) = x^3 + ax^2 + bx + c$ satisfies $p(1) = 0$, $p(-1) = 4$, and $p(2) = 15$. Find a, b, c .

3. Simplify: $\frac{x^3 - 8}{x^2 + 2x + 4}$.
4. Factorise: $a^3 + b^3 + c^3 - 3abc$. (State the identity from memory, then verify by choosing $a = b = c$.)
5. Given $a + b + c = 0$, simplify $\frac{a^3 + b^3 + c^3}{abc}$.
6. Prove that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ for $n = 1, 2, 3$ by direct verification, then use the formula to find $\sum_{k=1}^{10} (2k-1)^2$.
7. Simplify: $\frac{\frac{1}{a+b} - \frac{1}{a-b}}{\frac{1}{a+b} + \frac{1}{a-b}}$.
8. Make b the subject: $\frac{a-b}{a+b} = \frac{c-d}{c+d}$.
9. Given α, β, γ are roots of $x^3 - 7x + 6 = 0$, find $\alpha^2 + \beta^2 + \gamma^2$ and $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$.
10. Solve: $\frac{2}{x^2-1} - \frac{1}{x-1} = \frac{3}{x+1}$.

Section C Substitution & Structure

(≤90 s each)

One clean substitution is worth ten lines of algebra.

1. Solve: $\sqrt{x} - \frac{6}{\sqrt{x}} = 1$. (Let $t = \sqrt{x}$.)
2. Find the minimum value of $x^2 - 6x + y^2 + 4y + 20$ by completing the square in both x and y .
3. Show that $a^4 + b^4 \geq \frac{1}{2}(a+b)^4$ is **false**, then find and prove the correct inequality relating $a^4 + b^4$ to $(a^2 + b^2)^2$.
4. Solve: $x^4 - 2x^3 - 3x^2 + 4x + 4 = 0$ given that it has a repeated real root.
5. Given $a, b, c > 0$ with $a + b + c = 1$, prove that $ab + bc + ca \leq \frac{1}{3}$.
6. If $x + y = 1$, find the maximum value of $x^2y + xy^2$.
7. Solve over $\mathbb{R}$: $\sqrt{3x+1} - \sqrt{x+4} = 1$.

8. The equation $x^2 + bx + c = 0$ has roots r and s . For what values of b and c are the roots of $x^2 + cx + b = 0$ also r and s ?

Section D Challenge

(SMC / BMO1 difficulty)

No brute force. Each question rewards one key observation.

1. **Schur's Inequality (degree 2 case).** Prove that for all non-negative reals a, b, c :

$$a^2(a-b)(a-c) + b^2(b-a)(b-c) + c^2(c-a)(c-b) \geq 0.$$

(Consider the case $a \geq b \geq c \geq 0$ and group terms strategically.)

2. Find all polynomials $p(x)$ with real coefficients satisfying $p(x^2) = p(x)^2$ for all real x .

3. Evaluate:

$$\frac{1^2}{1 \cdot 3} + \frac{2^2}{3 \cdot 5} + \frac{3^2}{5 \cdot 7} + \cdots + \frac{n^2}{(2n-1)(2n+1)}.$$

Express your answer as a closed form in n .

4. A polynomial $p(x)$ of degree 3 satisfies $p(n) = \frac{1}{n}$ for $n = 1, 2, 3, 4$. Find $p(5)$.

5. Find all pairs of positive integers (m, n) with $m > n$ such that $m^2 - n^2 = 105$.

“In mathematics the art of proposing a question must be held of higher value than solving it.” — Georg Cantor

other prep to look at today:
Daily Integral for Calculus and Logic